How-To Library

Informatica

Configuring Intelligent Streaming 10.2 For Kafka on MapR

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Abstract

When you install Intelligent Streaming on MapR, you can run streaming mappings to read from any streaming source or write data to any streaming target without any additional configuration except for Kafka. This article provides the configuration steps that you need to perform to run streaming mappings to read from or write to Kafka on MapR.

Supported Versions

- Informatica Intelligent Streaming 10.2

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Overview

You can deploy Intelligent Streaming on Cloudera CDH, Hortonworks HDP, MapR, and Amazon EMR distributions.

You can create streaming mappings to stream data from sources, such as JMS providers, Apache Kafka brokers, Amazon Kinesis streams, and MapR streams and stream to targets, such as Kafka, HDFS, JMS providers, HBase tables, Hive tables, Amazon Kinesis Firehose, and MapR streams.

MapR Streams is a distributed messaging system that streams event data. It is integrated into the MapR Converged Data Platform and uses the Apache Kafka API. Although MapR Streams uses the Kafka programming model, there are differences in the implementation of Kafka.

When you create and run mappings on a MapR distribution, you can read from and write to MapR Streams and other streaming sources and targets except Kafka. To use Kafka data objects on MapR in streaming mappings, you need to perform configuration steps. The steps vary based on the version of Kafka that you deploy.

After you perform the configuration steps for Kafka, you cannot use MapRStreams data objects in streaming mappings on MapR.

Configuration Steps for Kafka 0.10.0.x or Earlier Versions

To use Kafka data objects of versions 0.10.0.x or earlier on MapR in streaming mappings, you need to perform configuration steps.

Perform the following steps:

1. Disable the Data Integration Service.
   When you disable the Data Integration Service, you shut down the service and disable the service process.
2. Back up the MapR distribution directory by making a copy of the directory.
3. Delete the rpm cache from the machine that hosts the Data Integration Service.
   The rpm cache is located in the following directories on the machine that hosts the Data Integration Service:
   - `<INFA_HOME>/tomcat/bin/<Temporary Directories>/infa_<dis_user>/<domain_name>/<dis_name>/`
   - `<INFA_HOME>/tomcat/bin/<Temporary Directories>/scalacache`
4. Update the Kafka version in the `hadoopRes.properties` file.

The `hadoopRes.properties` file is located in the following directory:

`<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/infaConf`

Include the following properties:

```
#Kafka version related properties
kafka.version=0.8.2.1
```

5. Copy the Kafka JAR files from the Cloudera distribution directory.
   Go to the following location:

   `<INFA_HOME>/services/shared/hadoop/cloudera_cdh5u10/lib`

   Copy the following JAR files:

   - kafka-clients-0.8.2.1.jar
   - kafka_2.11-0.8.2.1.jar
   - metrics-core-2.2.0.jar

   Place the files in the following directory on the Data Integration Service machine:

   `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/lib`

6. In the `hadoopEnv.properties` file, update the classpath with the new JAR files.
   The `hadoopEnv.properties` file is located in the following directory:

   `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/infaConf`

   Update the `infaspark.driver.cluster.mode.thirdparty.jars` and `infaspark.executor.thirdparty.jars` properties:

   a. Change

      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka_2.11-0.9.0.0-mapr-1611.jar,`**
      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka-clients-0.9.0.0-mapr-1611.jar`**

      to

      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka_2.11-0.8.2.1.jar,`**

      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka-clients-0.8.2.1.jar`**

   b. Add `$HADOOP_NODE_HADOOP_DIST/lib/metrics-core-2.2.0.jar` to both properties.

7. Delete the following JAR files:

   - `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/lib/kafka_2.11-0.9.0.0-mapr-1611.jar`
   - `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/lib/kafka-clients-0.9.0.0-mapr-1611.jar`

8. Start the Data Integration Service.
   You can now create and run a streaming mapping.

   Create a mapping to read from a Kafka source and write to any streaming target except MapRStreams.

Configuration Steps for Kafka 0.10.1.x or Later Versions

To use Kafka data objects of version 0.10.1.x or later on MapR in streaming mappings, you need to perform configuration steps.

Perform the following configuration steps:

1. Disable the Data Integration Service.
   When you disable the Data Integration Service, you shut down the service and disable the service process.
2. Delete the rpm cache from the machine that hosts the Data Integration Service.

The rpm cache is located in the following directory on the machine that hosts the Data Integration Service:

- `<INFA_HOME>/tomcat/bin/<Temporary Directories>/infa_<dis_user>/<domain_name>/<dis_name>/`
- `<INFA_HOME>/tomcat/bin/<Temporary Directories>/scalacache`

3. Update the Kafka version in the `hadoopRes.properties` file.
   The `hadoopRes.properties` file is located in the following directory:

   `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/infaConf`

   Include the following properties:

   ```
   #Kafka version related properties
   kafka.version=0.10.1.1
   ```

4. Copy the Kafka JAR files from the Cloudera distribution directory.
   Go to the following location:

   `<INFA_HOME>/services/shared/hadoop/cloudera_cdh5u10/lib`

   Copy the following JAR files:

   - kafka-clients-0.10.1.1.jar
   - kafka_2.11-0.10.1.1.jar
   - metrics-core-2.2.0.jar

   Place the files in the following directory on the Data Integration Service machine:

   `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/lib`

5. In the `hadoopEnv.properties` file, update the classpath with the new JAR files.
   The `hadoopEnv.properties` file is located in the following directory:

   `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/infaConf`

   Update the `infaspark.driver.cluster.mode.thirdparty.jars` and `infaspark.executor.thirdparty.jars` properties:

   a. Change

      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka_2.11-0.9.0.0-mapr-1611.jar`**`,`
      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka-clients-0.9.0.0-mapr-1611.jar`**

      to

      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka_2.11-0.10.1.1.jar`**`,`

      `$HADOOP_NODE_HADOOP_DIST/lib/`**`kafka-clients-0.10.1.1.jar`**

   b. Add `$HADOOP_NODE_HADOOP_DIST/lib/metrics-core-2.2.0.jar` to the properties.

6. Copy the `spark-streaming-kafka-0-10_2.11-2.0.2.jar` from the Cloudera distribution directory.

   Go to the following location:

   `<INFA_HOME>/services/shared/hadoop/cloudera_cdh5u10/lib`

   Copy the `spark-streaming-kafka-0-10_2.11-2.0.2.jar` file to the following location:
   `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/spark/jars`

7. Delete the following JAR files:

   - `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/lib/kafka_2.11-0.9.0.0-mapr-1611.jar`
   - `<INFA_HOME>/services/shared/hadoop/mapr_5.2.0/lib/ kafka-clients-0.9.0.0-mapr-1611.jar`

8. Start the Data Integration Service.
   You can now create and run a streaming mapping.

   Create a mapping to read from a Kafka source and write to any streaming target except MapRStreams.

Authors

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